Cryptography sheet # 7

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June 1, 2026

1 The notion of coset for groups

The notion of coset is the recurring idea of this course. We have the notion of coset for rings and we have it for groups, too. We really use this idea again and again. So, it makes sense to discuss the coset idea thoroughly. In Lagrange's theorem, we considered a group $(G, \cdot)$ written multiplicatively and introduced, for a given sub-group $H \subset$ G the cosets $gH = \{gh : h \in H\}$ of the group elements $g \in G$ modulo H. Here is an exercise that might help you digest this notion by considering a specific geometric example. A group of symmetries of a geometric shape is a group of invertible transformations that maps the shape onto itself. The group of rigid motions of a square that map the square onto itself is an example of such a group. We fix the square to have center in the origin and define it as the set of points (x, y) that satisfy $-1 \leq x, y \leq 1$. In that case, the square has the following group of the rigid-motion symmetries:

Formula	Pictogram	Word description
e(x,y)=(x,y)	$e(\boxed{\text{R}}) = \boxed{\text{R}}$	identity
a(x,y)=(-y,x)	$a(\boxed{\text{R}}) = \boxed{\rotatebox{90}{R}}$	90° degree counterclockwise rotation
b(x,y)=(-x,-y)	$b(\boxed{\text{R}}) = \boxed{\rotatebox{180}{R}}$	180° rotation
c(x,y)=(y,-x)	$c(\boxed{\text{R}}) = \boxed{\rotatebox{-90}{R}}$	90° degree clockwise rotation
h(x,y)=(-x,y)	$h(\boxed{\text{R}}) = \boxed{\reflectbox{R}}$	horizontal reflection
v(x,y)=(x,-y)	$v(\boxed{\text{R}}) = \boxed{\scalebox{1}[-1]{R}}$	vertical reflection
s(x,y)=(y,x)	$s(\boxed{\text{R}}) = \boxed{\rotatebox{90}{\reflectbox{R}}}$	north-west to south-east reflection
t(x,y)=(-y,-x)	$t(\boxed{\text{R}}) = \boxed{\rotatebox{-90}{\reflectbox{R}}}$	north-east to south-west reflection

This is a group $G = \{e, a, b, c, h, v, s, t\}$ with the group operation being the composition of the transformations. That is, $f \cdot g$ is defined to be the transformation $(f \cdot g)(x, y) = f(g(x, y))$. As you might notice, this group G is not Abelian. In the description of the group elements by pictograms, we just plotted a non-symmetric shape (the letter R) into the square, in order to trace the action of each group operation geometrically. But now, assume that we had some shape more symmetric thanR, to which we apply the group operations. For example, let us draw the letterAinto the square: $\boxed{\text{A}}$. This one is symmetric up to horizontal reflections. So, no matter whether we apply e or h to it, we obtain $\boxed{\text{A}}$. This gives us the subgroup $H = \{e, h\}$ of the symmetries of $\boxed{\text{A}}$ within the group G of the symmetries of $\boxed{}$. Your task is to determine all cosets gH where $g \in G$. This task has a clear geometric interpretation. When we apply the transformations in G to the shape $\boxed{\text{A}}$, we have 8 transformations, but only four possible outputs:

$$\boxed{\text{A}} \qquad \boxed{\rotatebox{90}{A}} \qquad \boxed{\rotatebox{180}{A}} \qquad \boxed{\rotatebox{270}{A}}$$

Each such output gives the coset of those transformations in G that send A to the respective shape. You can easily determine the cosets geometrically, but please also do the respective formula derivation, relying on the formal definition of the coset and the formulas for the elements of G.

If you like, you might also consider some other subgroup H' of G and calculate the cosets of G modulo H'.

2 Equivalence relation of Lagrange's Theorem

Let $(G, \cdot)$ be a group and H be subgroup of G. Show that the relation $\sim_H$ defined as:

$$\forall a, b \in G\colon \quad a \sim_H b \quad \Leftrightarrow \quad (\exists h \in H\colon \quad h \cdot a = b)$$

is a equivalence relation, i.e. prove that:

1. $\forall a \in G\colon \quad a \sim_H a$
2. $\forall a, b \in G\colon \quad a \sim_H b \Rightarrow b \sim_H a$
3. $\forall a, b, c \in G\colon \quad (a \sim_H b \ \wedge \ b \sim_H c) \Rightarrow a \sim_H c$

3 A Group of cosets

We know that $(\mathbb{Z}, +, \cdot)$ is a ring. After learning about the residue classes $\mathbb{Z}_n$ of $\mathbb{Z}$, we discovered that these also form a ring. For this, we simply reduced the addition and multiplication of residue classes to the addition and multiplication of representatives of these classes, i.e.:

$$\begin{aligned} \forall a, b \in \mathbb{Z}\colon \quad [a]_n + [b]_n &\coloneqq [a + b]_n \\ [a]_n \cdot [b]_n &\coloneqq [a \cdot b]_n \end{aligned}$$

Now one might ask whether a similar construction is possible with groups. Let $(G, \cdot)$ be a group and H a subgroup of G. For all cosets aH,bH, we want to define the operation $*$ as:

$$aH * bH \coloneqq (a \cdot b)H$$

The problem is that the above definition is somewhat ambiguous. There may exist $a, a' \in G$ and $b, b' \in G$ such that $aH = a'H$ and $bH = b'H$. (For example, if one chooses $G = (\mathbb{Z}_8, +), H = \{[0]_8, [4]_8\}$ then $[1]_8 H = \{[1]_8, [5]_8\} = [5]_8 H$). In this case,

$$aH * bH \coloneqq (a \cdot b)H \qquad \text{and} \qquad a'H * b'H \coloneqq (a' \cdot b')H$$

might contradict each other if $(a \cdot b)H \neq (a' \cdot b')H$.

(a) Show that this actually happens when we choose G and H like in exercise 1.

(b) Show that this cannot happen when G is Abelian by proving:

$$\forall a' \in aH : \forall b' \in bH : \quad (a' \cdot b')H = (a \cdot b)H$$

for all $a, b \in G$.

(c) Suppose G is Abelian and GH is set of all cosets of G regarding H. Is GH equipped with the above operation $*$ a group?